AUTOSAR High-Level Design (HLD) Document

Subsystem: Braking & Longitudinal Vehicle Dynamics Control | Version: 1.0

Document ID: HLD-AUTOSAR-BRAKE-2026-V1.0 | **Classification:** Automotive Confidential

1.0 System Architecture Overview & Scope

This High-Level Design (HLD) specifies the software architecture for the Next-Gen Braking and Longitudinal Vehicle Control system compliant with AUTOSAR Classic Release 4.4 and ISO 26262 ASIL-D functional safety standards. The architecture decomposes vehicle dynamics into specialized Software Components (SWCs) interacting via the AUTOSAR Runtime Environment (RTE).

1.1 Core Software Components Definition

The subsystem comprises the following key Software Components (SWCs):
• **BrakeController**: Application SWC (ASIL-D). Responsible for computing deceleration requests, hydraulic pressure modulation, and wheel slip regulation.
• **VehicleController**: Application SWC (ASIL-D). High-level vehicle supervisor coordinating propulsion, braking balance, and vehicle stability state.
• **SensorFusion**: Application SWC (ASIL-B). Consolidates perception telemetry from radar and wheel speed sensors into track estimates.
• **RadarInterface**: Sensor SWC (ASIL-B). Interfaces with front long-range radar hardware over CAN-FD.
• **SafetyController**: Complex SWC (ASIL-D). Independent watchdog and safe-state supervisor monitoring critical RTE signals.
• **CAN_Gateway**: BSW Module (QM). Manages cyclic CAN-FD frame communication with external ECUs.

2.0 Software Component Port Specifications

Each AUTOSAR Software Component exposes defined R-Ports (Required) and P-Ports (Provided) connected via RTE virtual functional buses.

Component (SWC)	Type	ASIL	Provided Ports (P-Port)	Required Ports (R-Port)
BrakeController	Application SWC	ASIL-D	P_BrakeCommand, P_BrakeStatus	R_VehState, R_SensorFusionTrack
VehicleController	Application SWC	ASIL-D	P_VehState, P_TargetTorque	R_DriverDemand, R_BrakeStatus
SensorFusion	Application SWC	ASIL-B	P_FusionTrack, P_ObstacleDist	R_RadarRaw, R_WheelSpeed
RadarInterface	Sensor SWC	ASIL-B	P_RadarRaw	R_CAN_RxMsg
SafetyController	Complex SWC	ASIL-D	P_SafeStateCmd	R_BrakeCommand, R_VehState
CAN_Gateway	BSW Module	QM	P_CAN_RxMsg	R_CAN_TxMsg

3.0 Interfaces and Signal Routing

The following Sender-Receiver and Client-Server interfaces are defined across SWC boundaries:
• **BrakeCommandInterface**: Source **BrakeController** -> Target **VehicleController**. Signals: TargetDeceleration (float32, m/s2), BrakePedalStatus (uint8), EBD_Active (boolean).
• **SensorTelemetryInterface**: Source **SensorFusion** -> Target **BrakeController**. Signals: ObstacleDistance (float32, m), RelativeVelocity (float32, m/s), TimeToCollision (float32, s).
• **RadarStreamInterface**: Source **RadarInterface** -> Target **SensorFusion**. Signals: RawTargetList (array of 16 targets), RadarStatus (uint8).
• **SafetySupervisionInterface**: Source **SafetyController** -> Target **BrakeController**. Signals: SafeStateActive (boolean), OverrideBrakeLevel (uint8).
• **VehicleTelemetryInterface**: Source **CAN_Gateway** -> Target **RadarInterface**. Signals: CAN_BusFrameData.

4.0 Functional Safety & ASIL Decomposition (ISO 26262)

Safety Goal SG-01: 'Prevent unintended deceleration or total loss of braking' is classified as ASIL-D. The BrakeController and SafetyController SWCs are allocated ASIL-D development lifecycle rigor. SensorFusion is assigned ASIL-B because environmental perception is validated against redundant driver override checks. CAN_Gateway operates at Quality Management (QM) with end-to-end (E2E) protection profiles implemented at the RTE level.

5.0 Execution Timing & Runnable Entities

- **BrakeController_Step (10ms Periodic)**: Executes anti-lock and deceleration PID algorithms.
- **VehicleController_Step (20ms Periodic)**: Coordinates powertrain and chassis arbitration.
- **SensorFusion_Step (20ms Periodic)**: Merges radar object tracks.
- **SafetyController_Watchdog (5ms Periodic)**: High-priority monitoring runnable.